

Regulatory semantic drift: when preference-based correction fails silently FINCAC40, a sixteen-year AMF corpus (2010–2026) for evaluating the temporal robustness of LLMs in ESG classification

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Abstract

Does an LLM trained on the standardized vocabulary of the CSRD (2023) recognize a sustainability issue formulated a decade earlier in a different register? This question of temporal robustness had remained untested for French documents, for lack of a multi-regime corpus sampled without bias toward recent documents.

For this first version, we present **FINCAC40**: 313,898 paragraphs from 23,144 AMF documents (CAC 40, 2010–2026), a Gold Standard of 140 ESRS-annotated paragraphs, and a stratified sampling protocol that avoids this bias. Across four LLMs, the result is heterogeneous: GPT-4o and Mistral-Large miss two out of three CSRD documents before 2015 but almost none after ($\Delta FNR = +0.54$ and $+0.45$), Mistral-7B exhibits the opposite trend, and Claude Sonnet 4.6 under-classifies almost everything, whatever the era. A linear probe of Mistral-7B confirms an above-chance temporal encoding (45.0% vs 30.7%), distributed over most layers. These results document an association between regulatory era and error rate for some of the models.

Finally, we document the failure, over two iterations, of a correction by preference optimization (ORPO): total collapse in the first run, then collapse onto a refuge category in the second (precision 5.3% on ESRS2, FPR 64.9%, failures degrading from 0 to 30.7%). In both cases, nothing in the training metrics (loss, accuracy) made it possible to detect the collapse: only a downstream, task-specific evaluation reveals it. **Correcting one bias through preference optimization can thus create another, more severe one, with no training signal to indicate it.** It is therefore a concrete case of shortcut learning under a constrained data regime. Corpus, annotations, code and artifacts, including those of the failure, are released in open access at <https://github.com/cdieng0/fincac>.

1 Introduction

1.1 Context and Motivation

Since the entry into force of the *Corporate Sustainability Reporting Directive* (CSRD, Directive (EU) 2022/2464) and of the *European Sustainability Reporting Standards* (ESRS, Delegated Regulation (EU) 2023/2772) in 2023, we have been witnessing a massive volume of publication of sustainability information structured according to a double materiality framework and coming from companies. These publications consist essentially of unstructured documents such as annual reports, regulatory press releases, or inside information disclosures, whose automated analysis is becoming an operational challenge for regulatory authorities, institutional investors and even audit firms.

Large language models (LLMs) are naturally considered as tools for automating this analysis, for tasks of extracting sustainability issues, classification according to the ESRS taxonomy, and

assessment of financial and impact materiality. This use implicitly rests on a rarely tested assumption: that current LLMs, trained mostly on corpora whose temporal distribution is dominated by the most recent data available at the time of *pretraining*, thereby possess a stable and uniform ability to recognize sustainability issues **independently of the period in which the analyzed document was written**.

However, this assumption is far from trivial. Indeed, the vocabulary of sustainability in the French language has undergone three major regulatory breaks between 2010 and 2026: the absence of a structured framework before 2015, the emergence of the French *Déclaration de Performance Extra-Financière* (DPEF) after the Grenelle II law and the Paris Agreement (2015), then the standardization imposed by the European Taxonomy (2020) and finally the CSRD (2023). Thus, one and the same reality — a company reducing its fossil energy consumption — may be described in 2011 in a generic vocabulary of “sustainable development” without reference to a standardized framework, and in 2024 in the precise vocabulary of the ESRS (“Scope 1/2/3”, “materiality”, or “climate transition plan”). If current LLMs associate the detection of a sustainability issue primarily with the presence of this recent technical vocabulary, they will systematically fail to identify the same issues expressed in an older register, which we call **temporal regulatory semantic drift**.

The existence of this phenomenon would have direct implications at the operational level, since an automated audit system based on an uncorrected LLM would produce systematic false negatives on historical archives, distorting any retrospective analysis of a company’s ESG trajectory.

1.2 The Empirical Gap

Despite the growth of the literature on LLMs applied to finance (Section 2), no study to our knowledge has empirically measured this phenomenon on a certified French regulatory corpus covering a period long enough to capture several distinct normative regimes. Existing English-language financial corpora (SEC EDGAR, FiQA) do not make it possible to study this question, for two reasons. First, US regulation relating to extra-financial sustainability is structurally different from, and less prescriptive than, the European framework. Second, most current financial NLP benchmarks are built in English, leaving the French language absent from the evaluation literature.

Moreover, building such an evaluation corpus raises a methodological difficulty that is rarely discussed explicitly in the literature. Indeed, selecting the documents to be annotated by relying on keyword filtering related to sustainability introduces a selection bias that would invalidate the intended measurement, since a CSRD keyword filter would mechanically favor recent documents (where this vocabulary is standardized) and under-represent older documents carrying the same semantic content expressed differently. This methodological difficulty in itself constitutes a contribution of our work (Section 3.4).

1.3 Research Hypothesis

The following central hypothesis is tested empirically in this work:

H1 (Exploratory hypothesis: regulatory recency bias) The CSRD classification performance of an LLM on a regulatory text is not independent of its period of writing. In other words, for equivalent sustainability semantic content, LLMs exhibit more false negatives on texts predating the standardization of the ESRS vocabulary.

Formally, denoting by $P_t(Y | X)$ the distribution of the sustainability category Y conditional on the text X as implicitly learned by the model for documents of period t , H1 predicts:

$$P_t(Y | X) \neq P_{t'}(Y | X) \quad \text{for } t \neq t'$$

Our hypothesis is however different from *covariate shift* in the classical sense (Shimodaira [12]), where $P(X)$ changes while $P(Y | X)$ is assumed stable. It is closer to a *concept shift* (or conditional *label shift*): the learned relation between text and category differs across eras. Our underlying, more precise assumption is that the actual semantic mapping between a sustainability issue and its category remains stable over time, but that X (the surface vocabulary) changes, and that the model relies excessively on lexical cues correlated with the period rather than on the underlying semantic content.

This hypothesis translates into a testable and falsifiable prediction, formulated cautiously given the sample sizes available per era (Section 3): the false negative rate (FNR) of an LLM evaluated on documents stratified by era should be higher on the older eras than on the recent ones. We do not predict strict monotonicity over only four points, but a clear contrast between the oldest end and the most recent end. The absence of such a contrast would constitute a refutation of H1 for the model considered.

1.4 Contributions

This first investigation combines an **empirical evaluation** contribution on the temporal robustness of LLMs on regulatory text, a **documented negative result** on the correction of this bias by preference optimization, and the **infrastructure** that makes these two contributions possible and directly reusable for other regulatory taxonomies. We present them in this order.

Contribution 1: A coherent representational signal on the temporal robustness of LLMs. We first test H1 through (i) a behavioral evaluation of the false negative rate stratified by era on four models covering two axes of comparison (scale effect within the Mistral family; generality of the phenomenon on two heterogeneous frontier models). Second, we carry out (ii) a linear probing of the internal representations of Mistral-7B, layer by layer. The results are mixed. Indeed, H1 is supported for two models out of four, contradicted for a third, while Claude Sonnet 4.6 exhibits a distinct profile, potentially related to its alignment rather than to a temporal effect. This heterogeneity shows that training and alignment influence the error profile of models faced with the same temporal shift. The linear probe provides a converging piece of evidence. In fact, temporal information is decodable above chance in most layers. These results nevertheless remain *descriptive and preliminary*. This is due to the small sample sizes per stratum, which do not allow statistical conclusions, and decodability does not demonstrate a causal link. They therefore justify a collaborative extension of the Gold Standard rather than a confirmation of H1.

Contribution 2: A documented and instructive failure of correction by preference optimization under a constrained data regime. Our attempts to correct the bias by ORPO all ended in failure. We observe a generalized collapse in the first run, then, after rebalancing, a collapse onto a “refuge” category (ESRS2, precision 5.3%). In both cases, no standard training metric made it possible to detect the collapse. We analyze the conditions of this failure (imbalance of pairs, fine-grained twelve-class taxonomy, absence of hard negatives for rare categories) and report them as a concrete case of shortcut learning (*shortcut learning* [6]) under a scarce-data regime, invisible to the training loop, a situation that is common in practice but little documented in the literature on *preference tuning*.

Contribution 3: A French regulatory corpus and a sampling protocol without selection bias. We build an entire extraction pipeline producing 313,898 paragraphs from 23,144 AMF regulatory documents, covering 16 years (2010–2026) and the current CAC 40 issuers. To

our knowledge, this is the largest French-language regulatory corpus available at this certified temporal depth. We associate with it a three-axis proportional stratification protocol (era, document type, CSRD density quartile) that explicitly avoids filtering on the variable of interest. Corpus, annotated Gold Standard, code and protocol are released in open access (Section 3.6), for reuse on other regulatory taxonomies or other questions of temporal robustness in finance and in NLP.

2 Related Work

The development of LLMs for finance rests essentially on two approaches: *continual pretraining* on financial corpora and *fine-tuning* on annotated texts. For these two approaches, the best-known models are respectively **FinGPT** (Yang et al. [15]) and **AdaptLLM / finance-LLM** (Cheng et al. [1]). These works, however, treat financial corpora as a relatively static domain and do not address regulatory evolution as a source of *distribution shift*.

This question connects more broadly to work on the temporal robustness of language models. The few related works existing to date are mainly concerned with **factual degradation** after training, also called *temporal misalignment*. It is in this sense that Lazaridou et al. [10] show that models trained on corpora cut off at a given date lose performance when they are evaluated on texts posterior to that date. A result that they explain by the appearance of new facts.

We will study the inverse phenomenon, termed **retrospective** drift, which consists in evaluating models on documents older than their knowledge period. The problem is therefore not the appearance of more recent information, but a **taxonomic** and **lexical** one, due to the evolution of the vocabulary used to describe the same content. A model trained mostly on data reflecting the standardized post-CSRD vocabulary thus risks failing to attach an older formulation to the corresponding semantic category.

This phenomenon is closely related to the problem of *covariate shift* in the sense of Shimodaira [12]. The latter showed that the distribution of formulations changes over time while the link between content and category remains stable. Work on **calibration drift** (Ovadia et al. [11]; Desai and Durrett [3]) has also shown the limits of neural networks when they are used out of distribution. Their confidence level then does not always correctly reflect their error rate. It is moreover this observation that motivated the initial inclusion of the *Expected Calibration Error* (ECE) among our metrics (Section 3.7).

3 Data and Methodology

3.1 Construction of the AMF Regulatory Corpus

3.1.1 Source and justification of the corpus choice

The corpus consists exclusively of documents from CAC 40 companies filed with the French **Autorité des Marchés Financiers (AMF)**, extracted from the public portal *info-financiere.gouv.fr*. This choice of a single source is certainly restrictive, but it is a deliberate decision, taken after evaluating several alternative sources.

Press articles such as **Les Échos / Yahoo Finance** introduce an editorial bias and a journalistic interpretation, whereas AMF documents are direct statements by issuers with the institutional vocabulary of the company and a certified AMF filing date. This temporal traceability is indispensable for studying the evolution of regulatory language over 16 years. In addition, since AMF

data are published under the Etalab Open Licence, their reuse and redistribution are authorized within the framework of this study.

From the source file (`flux-amf-new-prod.csv`, 454.1 MB, 524,589 entries since 2006), four filters were applied successively, the entries per stage of which are summarized in Table 1: keeping the current members of the CAC 40; then restriction to the French language; restriction to the 2010–2026 period; text extraction via `pdfplumber`, which fails on 2.56% of documents (404 errors or empty/scanned PDFs). In total, 23,144 documents are finally extracted, thus producing 313,898 paragraphs (Table 1).

Table 1: Statistics of the raw corpus by filtering phase

Phase	Entries
Raw AMF export	524,589
After CAC 40 filter	54,257
After FR language filter	30,263
After 2010–2026 filter	23,753
Successfully extracted (97.44%)	23,144
Paragraphs produced	313,898

Raw extraction produces text fragmented line by line (3 to 8 words per line), which makes a naive length filter ineffective: 95% of the extracts were rejected in our preliminary trials. We therefore reconstruct the paragraphs in two steps. The first step detects paragraph boundaries (empty lines, uppercase headings); the second merges wrapping lines up to strong punctuation, followed by filtering on structural criteria calibrated on 500 manually inspected paragraphs: length 20–340 words, at least 2 sentences, alphabetic ratio ≥ 0.55 (eliminates tables of figures) and unique-word ratio ≥ 0.30 (eliminates legal boilerplate). These criteria are structural, not semantic: a legal paragraph without ESG content satisfies them and remains in the corpus, where it will feed the *none* class of the taxonomy (Section 3.3).

3.2 Temporal regulatory regimes

Our central hypothesis is that regulatory discontinuities induce breaks in ESG vocabulary. We therefore define four eras corresponding to the major changes in the European and French normative framework for extra-financial disclosure, assigned by certified AMF filing date (Table 2). An annual granularity would produce strata that are too small ($n \approx 8$) on a Gold set of 140 examples; the four eras retained give $n \in [30, 42]$, sufficient to compute F1, FNR and Kappa per stratum with interpretable bootstrap confidence intervals.

Table 2: Definition of the four regulatory eras

Era	Period	Dominant regulatory characteristic	n Gold
2010–2014	Jan. 2010 → Dec. 2014	Absence of systematic ESG obligation; voluntary initiatives (GRI); informal vocabulary (“sustainable development”, “CSR”)	33
2015–2019	Jan. 2015 → Dec. 2019	Paris Agreement (2015); mandatory French DPEF (2017, art. 173); first structured CSR reports, without a European standard	30
2020–2022	Jan. 2020 → Dec. 2022	EU Taxonomy (2020); SFDR (2021); notion of double materiality emerging but not standardized	33
2023–2026	Jan. 2023 → Mar. 2026	CSRD in force (2023); ESRS Set 1 adopted; standardized vocabulary (“double materiality”, Scope 1/2/3, IRO, ESRS codes)	42

3.3 CSRD/ESRS Taxonomy

Our taxonomy takes up the ESRS Set 1 (Delegated Regulation (EU) 2023/2772): eleven thematic categories, plus a *none* category for texts outside the CSRD scope.

Of the 140 Gold annotations, 97 fall under *none*, i.e. 69.3%. This is not a sampling bias. Indeed, the AMF filings of the CAC 40 are for the most part share declarations, liquidity press releases or threshold crossings. This is why *none* always remains included before sampling (Section 3.4); removing it would give a corpus artificially centered on the CSRD, and therefore a model poorly calibrated in production.

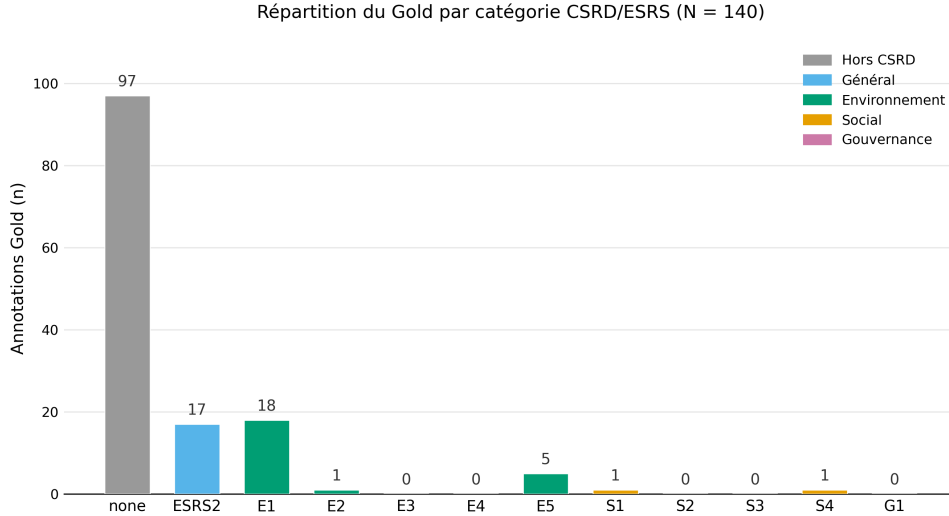


Figure 1: Distribution of the Gold set by CSRD/ESRS category

Figure 1 confirms the imbalance: E1 (climate) and ESRS2 (governance and strategy) concentrate the bulk of the CSRD mentions, whereas five standards E3, E4, S2, S3 and G1 do not appear in the Gold set.

Finally, the annotation follows the double materiality of the CSRD: an issue is financially material if it affects the company’s performance (*outside-in*), and impact-material if it affects people or the environment (*inside-out*). The two dimensions are annotated independently (Section 3.5).

3.4 Multi-LLM evaluation protocol

3.4.1 Model selection

We evaluate four models divided into two groups, each designed to isolate a different variable.

The first group tests the scale effect within a constant family: **Mistral-7B-Instruct-v0.3** (7 billion parameters, open-weights, natively French) against **Mistral-Large** (MoE architecture, undisclosed size, API, same family). Same tokenizer, same pretraining corpus, same dominant language — only size changes. If Mistral-Large drifts significantly less than Mistral-7B over time, this gain will be attributable to scale rather than to architecture.

The second group tests the generality of the phenomenon on two heterogeneous frontier models, both accessible via API: **GPT-4o** (OpenAI, predominantly English-speaking) and **Claude Sonnet 4.6** (Anthropic, non-French-speaking, strong RLHF alignment). Claude Sonnet 4.6 is

particularly interesting for observing the effect of RLHF alignment on a task with high regulatory stakes.

We note here, without making it a contribution of this work, that during the zero-shot screening that guided this selection, Claude Sonnet 4.6 exhibited a systematic under-classification with a profile distinct from the other models, potentially related to its alignment rather than to a temporal effect. The analyses of Section 4 concern exclusively Mistral-7B (the only open-weights model, a necessary condition for the probing of Section 4.2) and do not quantify this phenomenon; we mention it only as a lead, its rigorous characterization remaining future work.

We ruled out Llama and Qwen via OpenRouter: this third-party proxy does not make it possible to certify the exact version of the model used, which is problematic for reproducible academic work. A second Llama model, accessible directly via the Meta API, was also ruled out — Mistral-7B already covers the “small open-weights model” category, and adding it would only have made the protocol heavier without providing new information.

3.4.2 Prompt design

The system prompt is strictly identical for all models and all examples, a necessary condition for comparability. It details the twelve CSRD categories and the consistency rules of the *none* class, and imposes a strict JSON output with three fields: `csrd_category` (mandatory), `esrs_subcategory` (optional), and `chain_of_thought`, a three-step reasoning. This last field does not enter into any performance metric; it will serve for the qualitative analysis of errors and, later, for the construction of ORPO pairs.

All calls use a temperature $T = 0$ and a fixed seed (`random_seed = 42`) as far as the APIs allow. This configuration does not, however, guarantee strict determinism for proprietary models: the served model version, the serving infrastructure and the exact sampling implementation may vary without notice on the provider’s side. We systematically record the model identifier resolved by each call and keep the raw responses, which constitutes our guarantee of effective reproducibility rather than a theoretical guarantee of determinism. The native JSON mode (`response_format = {"type": "json_object"}`) is enabled for Mistral and OpenAI; Claude Sonnet 4.6, which does not offer this option, receives the JSON constraint only via the prompt.

3.4.3 Evaluation modes

In this part, we evaluate the LLMs following the zero-shot ($k = 0$) and 3-shot few-shot ($k = 3$) methods.

In zero-shot mode, we test the parametric knowledge of the models by injecting a system prompt that gives the taxonomy but no example.

In 3-shot mode, three examples that we annotated manually precede the classification instruction. They come from different eras and cover the three most represented categories (*none*, E1, ESRS2). They are then removed from the test set to avoid any data leakage. The test set in 3-shot mode therefore comprises 137 paragraphs ($140 - 3$) instead of 140.

The three examples retained each cover a target category: *none* (Danone, 2010-2014), E1 (STMicroelectronics, 2023-2026) and ESRS2 (STMicroelectronics, 2015-2019).

Important methodological limitation. This selection confounds, within the few-shot pool itself, category and era (*none*→old, E1→recent, ESRS2→intermediate). With 3-shot, a model could learn a surface heuristic linking the recency of the examples to their category rather than relying on the content of the paragraph. This contamination would bias precisely the measurement of temporal drift that the 3-shot condition aims to correct. An extension of this first version

would require testing several few-shot pools (fixed, randomized, balanced by era) and comparing the FNRs obtained. We have not done so in this version, and the 3-shot results must therefore be read as preliminary (Section 5).

3.5 Evaluation metrics

Five indicators measure the performance of the models from complementary angles: binary detection, overall agreement, bias by era, temporal drift, and inter-model comparison.

3.5.1 Binary F1 metric (CSRD vs. *none*)

We treat the detection of CSRD content as a binary classification: CSRD ($y = 1$, any category other than *none*) against non-CSRD ($y = 0$, category *none*). Since the corpus is imbalanced (69.3% of *none*), accuracy is misleading: a model that systematically predicted *none* would reach 69.3% of correct answers without any discriminative ability. We therefore use the binary F1 as the main metric, a standard measure introduced by van Rijsbergen [14] to combine precision and recall into a single score:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (1)$$

with TP the correctly identified CSRD documents, FP the *none* documents wrongly classified as CSRD, and FN the CSRD documents wrongly classified as *none*.

3.5.2 Cohen’s Kappa with bootstrap confidence intervals

Cohen’s Kappa κ [2] measures the agreement between the model’s predictions and the human annotations, corrected for chance agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (2)$$

where p_o is the proportion of observed agreement and p_e the agreement expected under independence:

$$p_e = \sum_c \frac{n_{c,\text{gold}}}{n} \cdot \frac{n_{c,\text{pred}}}{n} \quad (3)$$

With $n = 140$, parametric confidence intervals are not reliable. We estimate them by non-parametric bootstrap [5]: $B = 2,000$ resamplings (seed 42) for the global metrics, $B = 500$ for the per-era metrics, where the sample is smaller.

$$CI_{95\%}(\kappa) = \left[Q_{0.025} \left(\left\{ \hat{\kappa}^{(b)} \right\}_{b=1}^B \right), Q_{0.975} \left(\left\{ \hat{\kappa}^{(b)} \right\}_{b=1}^B \right) \right] \quad (4)$$

We interpret κ according to the Landis and Koch scale [9]: negligible agreement below 0.20, weak between 0.20 and 0.40, moderate between 0.40 and 0.60, substantial between 0.60 and 0.80, almost perfect above 0.80.

3.5.3 False negative rate by era (main indicator of H1)

The false negative rate (FNR) is a standard confusion-matrix quantity. Its use *stratified by era* to directly test a hypothesis of temporal drift, as well as the ΔFNR indicator below, are on the other hand constructions specific to this work. The FNR measures the share of the actual CSRD documents that the model wrongly classifies as *none*:

$$FNR(e) = \frac{FN(e)}{FN(e) + TP(e)} \quad (5)$$

for a given era e . This metric directly tests H1: if the models under-classify old CSRD documents as *none*, the FNR must decrease over time:

$$FNR(2010-2014) > FNR(2015-2019) > FNR(2020-2022) > FNR(2023-2026) \quad (6)$$

We define the FNR drift as:

$$\Delta FNR = FNR(2010-2014) - FNR(2023-2026) \quad (7)$$

A positive ΔFNR is consistent with H1; the higher it is, the more marked the apparent recency effect. Given the sample sizes per stratum, we treat this indicator as descriptive and not as a test.

3.6 Released artifacts and reuse conditions

Since the main contribution of this work is a resource, we specify here its content and conditions of use.

What we release. (i) The **FINCAC40** corpus: 313,898 paragraphs with, for each of them, the identifier of the source AMF document, the certified filing date, the issuer, the type of document and the derived regulatory era; (ii) the **Gold Standard** of 140 paragraphs annotated according to the ESRS taxonomy (category, and where applicable subcategory); (iii) the **code** for extraction, paragraph reconstruction, stratified sampling, evaluation and linear probing; (iv) the **training artifacts** of the two ORPO runs (preference pair sets, training logs, evaluation reports).

Where to find it. All of these artifacts are available in the repository <https://github.com/cdieng0/fincac>, which also contains the annotation protocol and the instructions needed to reproduce the evaluation, probing and ORPO runs reported in Section 4.

Licence and compliance. The AMF source documents are published under the Etalab Open Licence, which authorizes reuse and redistribution subject to attribution. The derived corpus and the code are distributed under a compatible permissive licence. The corpus contains only public regulatory documents of listed companies; it contains no personal data within the meaning of the GDPR, apart from the names of executives already appearing in the official publications.

Intended use and caveats. This resource is intended for the evaluation of models on French-language regulatory text and for the study of temporal robustness. The Gold Standard of 140 examples is a *preliminary* evaluation set: it is too small and too imbalanced (five ESRS categories absent) to serve as a training reference or to validate a compliance system in production. We explicitly invite the community to extend it, prioritizing rare categories and older eras, and we provide to this end the complete annotation protocol.

4 Results

This section illustrates the use of the corpus through three analyses: (1) a descriptive behavioral analysis of temporal bias in CSRD/ESRS classification, (2) an independent representational probe by linear probing, and (3) an attempt at correction by preference optimization (ORPO), whose documented failure constitutes a result in itself. The first two are preliminary in the sense defined in Section 3.

4.1 Behavioral temporal bias (H1)

On the 137 gold paragraphs outside the few-shot pool (Section 3), the four models are evaluated according to the protocol of Section 3: identical system prompt, zero temperature, fixed seed. We report the FNR (proportion of genuinely CSRD-relevant paragraphs wrongly classified as **none**) per time window, as well as $\Delta FNR = FNR(2010-2014) - FNR(2023-2026)$, which H1 predicts to be *positive*.

The result is heterogeneous across models (Table 3). The two frontier models go in the direction of H1, with a clearly positive ΔFNR : +0.537 for GPT-4o and +0.450 for Mistral-Large in zero-shot, these two models reaching a zero or near-zero FNR on the recent eras while missing two out of three CSRD documents on 2010–2014. Conversely, Mistral-7B shows a *negative* ΔFNR (−0.072 in zero-shot, −0.197 in 3-shot): its lowest FNR is observed on the *oldest* window, contrary to the prediction of H1. Claude Sonnet 4.6 is a distinct degenerate case: its FNR is close to 1 across all eras, the model classifying almost everything as **none** (cf. Section 3), which makes its ΔFNR uninterpretable as a temporal signal.

We therefore do not conclude that the data confirm H1. They support it for two models out of four and directionally contradict it for a third. One possible reading is that the recency effect manifests itself only in models performing well enough to exceed the difficulty floor of old documents: a model that massively under-classifies everywhere (Mistral-7B, and *a fortiori* Claude 4.6) cannot exhibit a temporal gradient, its FNR being saturated across all eras. This reading remains a conjecture: with 3 to 22 relevant paragraphs per window, no statistical test makes it possible to distinguish a real effect from sampling noise. We therefore consider these results as a *preliminary signal* justifying an extension of the Gold Standard.

Table 3: FNR per period and ΔFNR , four models, zero-shot and 3-shot. H1 predicts $\Delta FNR > 0$. Numbers of relevant (CSRD) paragraphs per window: 3 / 4 / 12 / 22. Since Claude Sonnet 4.6 classifies almost everything as **none**, its ΔFNR is not interpretable as a temporal signal.

Model	Mode	2010–14	2015–19	2020–22	2023–26	ΔFNR
GPT-4o	zero-shot	0.667	0.000	0.000	0.130	+0.537
	3-shot	0.667	0.000	0.000	0.136	+0.531
Mistral-Large	zero-shot	0.667	0.000	0.250	0.217	+0.450
	3-shot	0.667	0.000	0.333	0.364	+0.303
Mistral-7B	zero-shot	0.667	0.400	0.667	0.739	−0.072
	3-shot	0.667	1.000	0.750	0.864	−0.197
Claude S4.6	zero-shot	0.667	1.000	1.000	0.957	(−0.290)
	3-shot	1.000	1.000	1.000	1.000	(0.000)

The few-shot pool confounds era and category (Section 3). We therefore report both conditions and consider **zero-shot as the reference condition** for any reading of H1; the 3-shot

columns are provided for transparency, not as evidence. We note that the directional conclusions are unchanged between the two conditions for the three interpretable models.

For Mistral-7B (the only open-weights model, retained for the probing of Section 4.2 and the correction attempt of Section 4.3), the overall classification accuracy (12 categories) on the 137 paragraphs is 70.8 % and the macro F1 18.7 %. This gap is due to the class imbalances of the corpus (`none` being largely dominant) rather than to a homogeneous performance per category.

4.2 Representational evidence (linear probing)

To test whether temporal information is encoded in the internal representations of the model independently of its output behavior, we extracted the hidden states of the last token for each of the 33 layers of Mistral-7B-Instruct-v0.3 (embeddings + 32 transformer blocks) on the 140 gold paragraphs, and trained an L2-regularized multinomial logistic regression ($C = 0.1$) per layer to predict the era (4 classes) from this representation alone, with stratified 5-fold cross-validation.

Layer 19 obtains the best accuracy (45.0 %, against a majority baseline of 30.7 % and chance at 25.0 %). A permutation test (200 draws) gives $p = 0.005$ for this layer. This value, however, corresponds to the minimal resolution threshold of the test and is not corrected for multiple comparisons over the 33 layers (cf. Limitations, point 2).

The signal is nevertheless not an isolated artifact on a single layer: indeed, almost all of layers 5 to 32 exceed the majority baseline in a stable manner (Figure 2), which indicates a distributed encoding of temporal information rather than a one-off peak. This result contradicts the localized linguistic pipeline (POS, syntax, semantics, coreference across successive layers) that Tenney et al. [13] report on BERT.

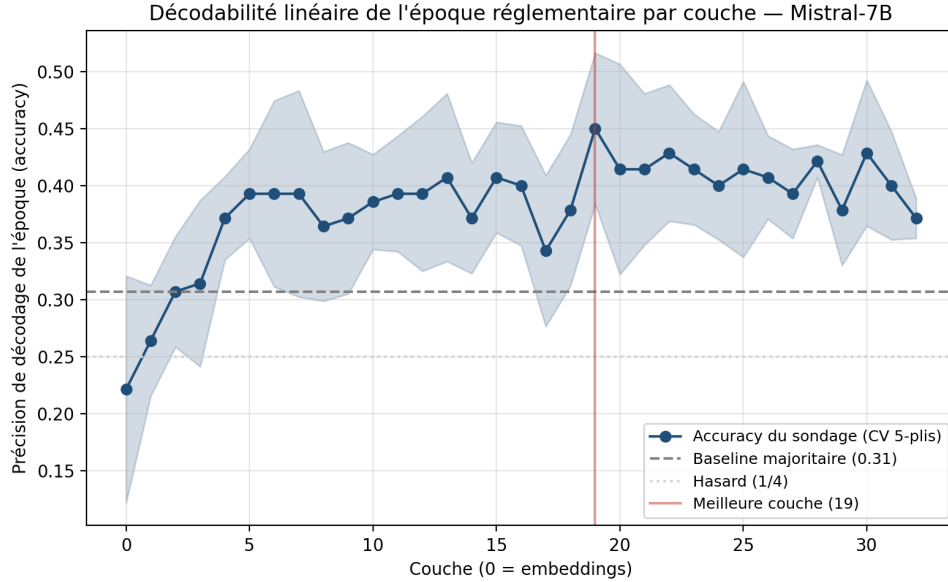


Figure 2: Linear decodability of the regulatory era per layer — Mistral-7B. The vertical line marks the best layer (19). The shaded area represents the standard deviation over the 5 cross-validation folds.

4.3 Attempted correction by ORPO: an informative negative result

To attempt to correct the bias documented in 4.1, we collected preference pairs by *rejection sampling* on unannotated paragraphs of the corpus (outside the gold set, exclusion verified by ID and by content), comparing the predictions of a strong judge (**mistral-large-latest**) with those of the base policy (**open-mistral-7b**, 3-shot conditions identical to Section 4.1). The labels produced by the judge are *silver labels*, not an independent ground truth: judge and policy belong to the same family of models, which limits the scope of the correction to what **mistral-large-latest** “knows” how to identify and does not exclude a bias shared between the two models (Section 5). The disagreements were categorized into three types: A) recency false negative (silver = CSRD, base = *none*), B) over-prediction false positive (silver = *none*, base = CSRD), C) category confusion, then used for a fine-tuning in 4-bit quantization (QLoRA [4], adapters of rank $r = 16$, $\alpha = 32$ according to the LoRA parameterization [8]) combined with the ORPO preference objective [7] ($\beta = 0.1$).

Run 1. On 300 A / 10 B / 5 C (273 training pairs, 3 epochs, $lr = 5 \times 10^{-5}$), training converges normally from the point of view of the ORPO objective (preference accuracy on validation: 100 % at the end of training) but the corrected model collapses on the actual classification task (Table 4): the zero FNR across all eras does not signal a correction of the bias, it results from a generalized calibration collapse. In fact, the model stopped predicting *none*, which mechanically cancels the FNR without correcting the cause. We attribute this collapse to the extreme imbalance between pair types (ratio $\sim 30:1$ in favor of A) combined with the absence of pairs explicitly reinforcing that a correct *none* prediction must be preserved.

Run 2. After capping the A vs. guardrail (B+C) ratio at 5:1 and oversampling the guardrail pairs $\times 3$ (225 A / 30 B / 15 C), as well as fixing a context-budget truncation bug and cautiously lowering the learning rate, accuracy improves but remains low (Table 4). Adding the false positive rate to our evaluation protocol reveals a global FPR of 64.9 % that is almost constant across the four eras (0.70 / 0.65 / 0.67 / 0.55): the model still massively over-predicts CSRD, largely independently of the era of the document. A fine-grained inspection reveals a more specific mechanism than a generalized bias: the **ESRS2** category simultaneously exhibits a precision of 5.3 % (1/19) and a recall of 5.9 % (1/17), a sign that it has become a *refuge* category triggered in particular by the generic term “risk” in standardized legal clauses, without discrimination from a real sustainability issue.

Table 4: Classification on the gold-140, base policy and two ORPO correction runs. Binary κ (CSRD vs. *none*), 95% CI by bootstrap ($B = 2,000$, seed 42, protocol of Section 3). “Parsing fail.”: proportion of responses whose JSON is unreadable, counted as absence of a detected category for the computation of κ and as errors for accuracy and macro F1. The κ of Run 1 remains unavailable, as its individual predictions were not preserved.

	Accuracy	Macro F1	κ [95% CI]	Parsing fail.	FNR (per era)	Global FPR
Base policy (3-shot)	70.8 %	18.7 %	0.193 [0.051, 0.352]	0 %	0.667–1.000	n/a
ORPO — Run 1	16.4 %	23.1 %	n/a	n/a	0.0	n/a
ORPO — Run 2	35.7 %	35.9 %	0.238 [0.079, 0.387]	30.7 %	0.0–0.043	64.9 %

κ goes from 0.193 for the base policy to 0.238 for Run 2, i.e. an agreement that remains *weak* in both cases according to the Landis and Koch scale [9]. Above all, the confidence intervals overlap widely ([0.051, 0.352] against [0.079, 0.387]): this apparent progression is not distinguishable from sampling noise on $n = 140$, and we do not claim it as an improvement. To avoid any ambiguity:

Run 1 is a total collapse (the model stopped predicting *none*), and Run 2, despite a numerical gain in accuracy, remains a clear failure of the intended correction. We note that an FPR of 64.9 %, a refuge category with 5.3 % precision, and a κ statistically indistinguishable from that of the uncorrected model do not constitute a correction, even partial, of the targeted bias.

To this is added a degradation that our initial evaluation protocol masked: the model resulting from Run 2 produces an unreadable JSON output in **30.7 %** of cases, against 0 % for the base policy. The *fine-tuning* therefore not only failed to correct the temporal bias; it degraded the model’s ability to comply with the imposed output format, an ability that the base model mastered perfectly. This observation reinforces the interpretation of a shortcut learning rather than of a skill acquisition.

Why preference-based correction fails here. Four factors, cumulative rather than alternative, explain this failure: (i) an extreme imbalance between pair types in Run 1, partially corrected in Run 2; (ii) a fine-grained twelve-category taxonomy, whereas 273–315 pairs rarely suffice to cover the diversity of cases necessary for a preference-based correction over such an output space; (iii) the absence of high-quality *hard negatives* for rare categories, since *rejection sampling* produces pairs only where judge and policy diverge, which mechanically under-samples the categories that are already poorly represented in the gold set; (iv) a judge and a policy from the same model family, which does not guarantee the absence of a shared bias (point above). We attribute the failure to this constrained data regime and to this combination of factors, not to an intrinsic limitation of preference optimization or of ORPO as a method.

Notable methodological point: nothing in the standard training metrics (eval_loss, preference accuracy) made it possible to detect these collapses. ORPO evaluates itself on its own relative preference task, not on the downstream classification task, nor on the false positive rate, nor on the validity of the output format. A periodic evaluation callback on a gold subset, explicitly measuring FNR, FPR *and* parsing failure rate during training, would have allowed an early detection from Run 1 onwards.

5 Limitations

1. **Semantic equivalence between eras not established.** H1 implicitly assumes that old and recent documents dealing with the same sustainability issue are semantically comparable, only the surface vocabulary differing. Our protocol does not verify this equivalence: the difference in FNR observed between eras may reflect, in whole or in part, factors confounded with the period rather than a purely lexical drift (length of the documents, distribution by document type or by company, etc.). This is the most important limitation of this work. Controlled experiments would be necessary to isolate it: (i) classification after removal of the explicit regulatory markers (“CSRD”, “ESRS”, “Scope 1/2/3”, “double materiality”); (ii) paraphrase neutralizing the temporal register; (iii) counterfactuals crossing old content with modern vocabulary and vice versa. We have not conducted them in this version and therefore present our results as an association observed between era and error rate, not as a causal demonstration of *semantic drift*.
2. **Survivorship bias in the composition of the corpus.** The CAC 40 filter (Section 3) retains the issuers *currently* members of the index, then collects their documents over 2010–2026. A company that has left the index in the meantime is not represented, even if it was an important issuer over part of the period. The corpus therefore documents the regulatory trajectory of the companies that *persisted* in the CAC 40 until the construction of the

dataset, not the complete universe of large French capitalizations over 16 years.

3. **Restricted gold sample ($n = 140$).** The FNRs per era (Section 4.1) rest on numbers of 3 to 22 relevant paragraphs per time window. This result is insufficient to establish a strictly monotonic trend with statistical confidence, as discussed in Section 4.1. Five ESRS categories (E3, E4, S2, S3, G1) are absent from the gold set and two others (E2, S4) count only a single example each; in practice, our multiclass evaluation largely reduces to a binary CSRD / *none* detection on the dominant categories (E1, ESRS2). An extension of the gold set would be necessary for finer per-category conclusions.
4. **Possible confusion between era and few-shot pool.** As noted in Section 3, the three examples of the 3-shot prompt confound category and era, which may inject a surface shortcut into the 3-shot results independently of any real temporal effect of the model.
5. **Layer selection and multiple comparisons (probing).** Layer 19 was chosen as the best performance among 33 layers tested, then tested for its significance. A procedure subject to the classical problem of multiple comparisons. The reported p -value (0.005) is not corrected (Bonferroni over 33 layers would give an effective threshold ≈ 0.0015 , not reached) and corresponds in any case to the resolution floor of the permutation test (200 draws). The conclusion therefore rests more on the robustness of the plateau observed over layers 5–32 (Figure 2) than on the point figure of layer 19. Moreover, this signal could at least partially reflect explicitly temporal lexical markers (dates, standard names) rather than a more abstract representation of the era. A confusion that only the control experiments of point 1 would make it possible to settle.
6. **ORPO correction not completed despite two iterations, with a judge from the same family as the corrected policy.** Run 1 identified and corrected the extreme imbalance of pair types; Run 2, corrected accordingly, improves accuracy (16.4 % $\rightarrow$ 35.7 %) but reveals a second, distinct cause that we did not have the opportunity to correct and validate with a third run. Moreover, the silver judge (`mistral-large-latest`) and the corrected policy (`open-mistral-7b`) belong to the same model family: a bias shared between the two is not excluded, and a human validation of a sample of the silver labels, or a judge from a different family, would strengthen the credibility of the correction. The fact that one corrected cause reveals a second, distinct one further suggests that preference-based correction of this bias requires a volume of data per category greater than what our current gold set allows. Finally, since the individual predictions of Run 1 were not preserved, its κ and any paired test (e.g. McNemar) between the three configurations remain out of reach; we have since corrected the evaluation protocol to systematically archive the individual predictions and the parsing failure rate, whose initial underestimation had masked a 30.7 % degradation in the corrected model’s compliance with the output format.
7. **Causal link between representational and behavioral bias not established.** The probing (Section 4.2) demonstrates that temporal information is linearly decodable from the internal representations, but does not demonstrate that the model actually *uses* this signal for its downstream classification decisions. Establishing this causal link (for example via activation patching) remains a direction for future work and is beyond the scope of this preprint.
8. **Annotation by a single expert,** without a measure of inter-annotator agreement on the gold-140 (contrary to the protocol planned for the future 500-example hold-out set).

6 Conclusion

This work delivers a research infrastructure for the study of the temporal robustness of LLMs on French-language regulatory text: **FINCAC40**, a corpus of 313,898 paragraphs from 23,144 AMF documents covering sixteen years and the *current* CAC 40 issuers (survivorship bias discussed in Section 5), together with an annotated Gold Standard, a sampling protocol that explicitly avoids selection bias on the variable of interest, and all the code and artifacts (Section 3.6). It is this resource, more than the analyses it enables at this stage, that constitutes our main contribution.

We frame this work as an investigation rather than a demonstration: do classification performances vary with the era of the document? Is temporal information decodable from the internal representations? These two questions receive a positive answer, but do not suffice to establish a causal semantic drift in the strong sense. On the base policy studied in depth (Mistral-7B, 3-shot), the false negative rate remains high across all four regulatory regimes without strictly monotonic progression. A result compatible with a recency effect superimposed on a more general under-classification bias, not separable with the available sample sizes ($n = 140$), and that we cannot distinguish from a set of confounded factors (length, document type, company, topic frequency) for lack of having established the semantic equivalence between old and recent documents (Section 5, point 1). The linear probing of the internal representations provides an independent piece of evidence consistent with this behavioral observation: temporal information is linearly decodable there with an accuracy above chance, distributed over most layers rather than localized in a single detector. Through this probe we have established a correlational decodability. We also note that this signal could partially reflect explicitly temporal lexical markers rather than a deeper abstraction of the notion of era.

Our attempt at correction by preference optimization documents an instructive failure over two iterations: the first reveals a generalized calibration collapse, correctable by rebalancing the pair types; the second, after this correction, reveals the spontaneous emergence of a refuge category absorbing ambiguous cases. We choose to document this failure in detail rather than omit it: the volume of data necessary for a reliable preference-based correction on a twelve-category taxonomy manifestly exceeds what a modest annotation budget allows to collect, a constraint that we consider more representative of common practice than the abundance of data assumed by the reference literature on these methods.

The entire corpus, the annotations, the evaluation and probing code, as well as the complete artifacts of the two ORPO correction attempts, are released in open access at <https://github.com/cdieng0/fincac>, with the aim of constituting a starting point reusable by researchers in the field for the study of the temporal robustness of LLMs on specialized regulatory taxonomies.

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